

tests/regression/ external_egress_verdict_test.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T07:15:00Z **Status:**
Active standing regression guard (§11.4.135) for BUGFIX-0012.

Companion (§11.4.18) to the in-source documentation block at the top of the script. Sibling of docs/scripts/comprehensive_admin_topology_test.md.

Overview

A standing regression guard that proves tests/comprehensive-test.sh never reports a proxy **FAIL** for a **third-party outage**, and never masks a **real proxy defect** as an outage **SKIP**.

Pre-fix, the "various sites" loop in test_http_proxy() and the concurrency test in test_concurrent() decided purely on proxy_http_code == 200 ? PASS : FAIL. When an external endpoint (httpbin.org) was **down**, the direct host fetch failed too — yet the suite hard-**FAILED** on an outage the proxy did not cause: a §11.4.1 false-FAIL that made the suite non-deterministic (§11.4.50) and not re-runnable (§11.4.98). The working sibling api.ipify.org PASSED at the same moment, proving the proxy's egress was fine.

The fix routes both call sites through a shared pure classifier _external_egress_verdict(proxy_code, direct_code):

proxy_code	direct_code	verdict	meaning
200	(any)	PASS	the proxy fetched it
≠ 200	200	FAIL	the proxy cannot fetch a site the host reaches directly — a real proxy defect (the anti-bluff catch)
≠ 200	≠ 200	SKIP	the endpoint is down for everyone — external outage, not ours (§11.4.3, network_unreachable_external)

Same discipline the sibling `test_large_file()` already uses for its faithful-relay / network_unreachable_external gate.

It does **not** grep the source. It extracts the **real** `_external_egress_verdict` function and drives it with the three canonical code pairs — deterministic, no network.

Prerequisites

- POSIX sh, awk, mktemp. No network, no privileges.

Usage examples

```
# GREEN guard (default) – assert PASS(200,.) + FAIL(≠200,200) +
  SKIP(≠200,≠200):
tests/regression/external_egress_verdict_test.sh
  # exit 0 = PASS

# RED reproduce – the pre-fix (proxy!=200 => FAIL) replica FAILs
  an outage pair:
RED_MODE=1 tests/regression/external_egress_verdict_test.sh
  # exit 0 = defect reproduced

# Runs automatically inside the suite:
bash tests/run-tests.sh                                     #
  test_regression_guards()
```

§11.4.115 RED_MODE polarity

RED_MODE	What it runs	PASS means
0 (default)	the REAL _external_egress_verdict from comprehensive-test.sh	(200,000)→PASS and (503,200)→FAIL and (503,000)→SKIP → the outage is SKIPped (bluff refused) and a real defect still FAILs (GREEN guard)
1	the PRE-FIX replica (proxy!=200 ⇒ FAIL)	(503,000)→FAIL → the false-FAIL on an external outage reproduces

A RED_MODE=1 run that cannot reproduce is a finding per §11.4.7, not a pass.

Edge cases

- **Fix reverted** (`_external_egress_verdict` flipped back to proxy!
`=200 ⇒ FAIL`) → the GREEN guard reports REGRESSION: verdict
wrong ... and exits 1 (the §1.1 paired-mutation behaviour,
verified against the real function).
- **Fail-open regression** (outage → PASS instead of SKIP, or real
defect → SKIP) → the GREEN guard's
`OK200=PASS*DEFECT=FAIL*OUTAGE=SKIP` pattern no longer matches
→ FAIL.

Internal behaviour

- `#!/bin/sh, set -eu; POSIX-only, sh -n parse-clean (§11.4.67).`
- Extraction: `awk '/^_external_egress_verdict\(\\) \/{f=1} f{print} /^\\}/{if(f){exit}}'`.
- Writes one evidence file per run under `qa-results/regression/
external_egress_verdict/` (gitignored).

Related

- Fix site: `tests/comprehensive-test.sh` —
`_external_egress_verdict()`, the sites loop in `test_http_proxy()`,
the direct pre-probe in `test_concurrent()`.
- Sibling gate: `test_large_file()` `network_unreachable_external`
SKIP.
- `tests/run-tests.sh` — registers this guard via
`test_regression_guards()`.
- `docs/issues/fixed/BUGFIXES.md` — BUGFIX-0012.

Last verified

2026-07-01 — `sh -n parse-clean`; GREEN asserts PASS/FAIL/SKIP
for the three canonical pairs (exit 0); `RED_MODE=1` reproduces the
false-FAIL (exit 0); paired mutation (revert the helper) makes
GREEN FAIL (exit 1) then restores byte-identical; live
`comprehensive-test.sh` = 33 pass / 0 fail / 10 skip with `httpbin.org`
down.